Multi-Source Evidence Fusion and Calibrated Probabilistic Networks for Legacy Petrol Engine Diagnostics

Executive summary

The diagnostic engine under study, which operates on legacy petrol passenger cars (model years ≈ 1990–2010), faces a diagnostic performance plateau with top-1 accuracy limited to $50–52\%$. Analysis indicates that the current hand-tuned, additive, weighted "prune-then-rank" classifier is the primary system constraint. This linear, deterministic architecture is incapable of resolving non-linear diagnostic spaces, conditional sensor dependencies, or severe sensor contradictions.

A systematic review of cross-domain benchmarks reveals that hand-tuned rule bases suffer an absolute performance penalty of $15–35\%$ compared to learned statistical classifiers and Bayesian networks[1]. To overcome this constraint without discarding valuable expert knowledge, a transition to a calibrated probabilistic fusion model is required. This report demonstrates that:

1. Additive rule engines have an inherent mathematical ceiling that prevents scalable multi-class diagnostics under noisy or missing data conditions[3].
2. Sensor contradictions can be modeled as active diagnostic features using Dezert-Smarandache Theory (DSmT) or Proportional Conflict Redistribution (PCR5/6) to prevent Zadeh's paradox[4].
3. Conditional dependencies between live sensor data and exhaust chemistry must be resolved via Stacked Bayesian Networks or Log-Odds Aggregator frameworks to prevent "swamping" reliable channels[6].

Rule-based vs learned rankers: the achievable ceiling

The first research question addresses the performance limit of hand-tuned rule-based classifiers relative to learned statistical rankers on multi-class diagnostic tasks with noisy, partially missing evidence.

Finding

Hand-tuned rule engines and additive weight rankers consistently plateau at top-1 accuracies of $50–62\%$ on complex multi-class diagnostic tasks involving noisy or incomplete data[3]. Under identical data conditions, learned statistical and probabilistic classifiers—including Bayesian networks, Support Vector Machines (SVMs), and Random Forests—routinely achieve top-1 accuracies of $80–99\%$ [1]. The absolute performance gap between hand-tuned rule

systems and learned approaches ranges from 15% to over 35% [1].
The primary ceiling for the current "prune-then-rank" engine is its linear additive structure. It cannot resolve non-linear diagnostic boundaries where symptoms overlap, nor can it handle cases where diagnostic trouble codes (DTCs) are absent, which starves the rule base of its primary triggers[3].

Evidence and sources

The most rigorous head-to-head comparison between a hand-tuned rule-based expert system and a Bayesian network on identical noisy datasets is documented in the HEPAR medical diagnostic project[3]. HEPAR-RB is a hand-crafted rule-based expert system utilizing 178 hierarchical production rules[3]. HEPAR-BN (and its successor HEPAR II) is a causal Bayesian network containing over 70 nodes and conditional probability tables (CPTs) learned from clinical cases[3]. Both models were evaluated on the IFF dataset (699 patients) and the DH dataset (181 patients) to diagnose complex, overlapping disorders under conditions of missing symptoms and laboratory findings[3].
The diagnostic results for Primary Biliary Cirrhosis (PBC) are detailed in the table below[3]. In this evaluation, "Unclassified" (UC) designates cases where the rule base failed to activate or the Bayesian network did not clear a conservative 50% probability threshold[3].

Evaluation Dataset & Class	**Hepar-RB (Rule-Based Expert System)**	**Hepar-BN (Bayesian Network)**	**Absolute Performance Gap**
IFF Dataset (699 Cases)			
Actual Positive PBC (280 cases) Correct	62.0%	94.0%	$+32.0\%$ **(BN Superiority)**
Actual Positive PBC Unclassified (UC)	34.0%	1.0%	-33.0% (BN Red. Uncertainty)
Actual Negative PBC (419 cases) Correct	20.0%	45.0%	$+25.0\%$ **(BN Superiority)**
Actual Negative PBC Unclassified	76.0%	40.0%	-36.0% (BN

(UC)			Red. Uncertainty)
DH Dataset (181 Cases)			
Actual Positive PBC (15 cases) Correct	73.3%	80.0%	$+6.7\%$ **(BN Superiority)**
Actual Positive PBC Unclassified (UC)	13.3%	0.0%	-13.3% (BN Red. Uncertainty)
Actual Negative PBC (166 cases) Correct	86.0%	39.0%	-47.0% (RB Specificity Edge)
Actual Negative PBC Unclassified (UC)	13.0%	60.0%	$+47.0\%$ (BN Conservative Hold)

In the larger and noisier IFF dataset, HEPAR-RB left 76% of negative cases and 34% of positive cases unclassified due to rule starvation[3]. The Bayesian network resolved these cases by calculating posterior probabilities over the remaining weak signals, achieving a 94% sensitivity on positive cases[3]. This shows how a probabilistic framework handles missing inputs without stalling.

In the engineering domain, similar performance gaps are documented. In multi-class fault classification across twelve distinct grid faults, SVMs, Decision Trees, and LSTMs achieved accuracies of $96–98\%$ [1]. Conversely, traditional rule-based systems struggle with rule explosion and poor generalization when operating outside of tightly controlled, stable environments[11].

In aluminum electrolysis cells, an optimized rule-based system utilizing certainty factors achieved 94.2% diagnostic accuracy, which matched an LSTM model (93.8%) but required extensive manual tuning of disjunctive and conjunctive certainty rules[2].

Confidence

High. The HEPAR benchmark represents a long-term, peer-reviewed evaluation of rule-based

and Bayesian networks on identical, noisy real-world data[3].

Data gaps and domain-transfer caveats

The HEPAR dataset represents a medical domain[3]. However, the domain transfer to automotive fault diagnosis is structurally valid. Medical findings (patient symptoms, physical signs, and blood tests) correspond to the four automotive diagnostic channels under study: operator symptoms, live sensor telemetry, 5-gas exhaust chemistry, and DTC presence.
Both systems map noisy, incomplete, and conditionally dependent features to a discrete, mutually exclusive or near-exclusive label space. Unlike human physiology, automotive systems are bounded by rigid thermodynamic and chemical equations (such as Stoichiometric combustion chemistry), which should make a probabilistic physical structure more robust than its medical counterpart.

Contradiction as a discriminator without destructive override

The second research question examines principled mathematical methods that allow a classifier to use evidence contradiction as an active diagnostic feature, rather than as a hard, destructive override.

Finding

When independent evidence channels contradict each other—such as when live ECU sensor data indicates a stoichiometric mixture ($\lambda \approx 1.0$) but the 5-gas exhaust analyzer measures high oxygen levels ($O_2 > 2\%$) indicating a lean burn—traditional rule-based systems rely on hard overrides. These overrides are highly sensitive to single-channel failures, often resulting in diagnostic errors.
In evidence theory, classical Dempster's rule of combination fails under high contradiction ($K \to 1$), leading to overconfident, counter-intuitive conclusions known as Zadeh's paradox[4].
To constructively utilize contradiction without destructive overrides, modern multi-sensor fusion architectures employ Proportional Conflict Redistribution (PCR5 or PCR6) rules[4]. These methods mathematically isolate the conflicting probability mass and redistribute it locally to the specific channels generating the contradiction, preserving the semantic meaning of the disagreement[4].

Evidence and sources

The hazard of classical evidence fusion under contradiction is demonstrated by Zadeh's paradox[4]. Let a diagnostic system evaluate three mutually exclusive fault families: Rich mixture (R), Lean mixture (L), and Mechanical wear (W). Let the basic belief assignments (BBAs) from two independent sensors be:

$$m_1(R) = 0.99, \quad m_1(L) = 0.01, \quad m_1(W) = 0.00$$

$$m_2(W) = 0.99, \quad m_2(L) = 0.01, \quad m_2(R) = 0.00$$

Here, the first channel indicates a Rich mixture with 99% confidence, while the second channel indicates Mechanical wear with 99% confidence. They weakly agree on a Lean mixture (1%)[13].

Under classical Dempster-Shafer Theory, the total conflict mass K represents the sum of the products of conflicting evidence[13]:

$$K = \sum_{A \cap B = \emptyset} m_1(A) m_2(B) = m_1(R)m_2(W) + m_1(R)m_2(L) + m_1(L)m_2($$

$$K = (0.99 \times 0.99) + (0.99 \times 0.01) + (0.01 \times 0.99) = 0.9999$$

The denominator normalization factor becomes $1 - K = 0.0001$ [4]. Applying Dempster's combination rule, the joint belief for the weakly supported Lean mixture L is[13]:

$$m(L) = \frac{m_1(L)m_2(L)}{1 - K} = \frac{0.01 \times 0.01}{0.0001} = 1.00$$

This yields a 100% certain diagnosis of a Lean mixture[13]. This result is highly counter-intuitive and dangerous for a physical engine; a minor agreement in a high-conflict scenario completely overrides the dominant diagnostic signals[4].

To prevent this failure mode, alternative combination rules have been developed. The table below compares how different fusion methods distribute probability mass under this

high-conflict scenario[5]:

Fusion Operator / Method	Mass Assigned to R (Rich)	Mass Assigned to W (Wear)	Mass Assigned to L (Lean)	Mass Assigned to Ignorance / Unknown
Dempster's Rule (Classic DST) [cite: 14]	0.000	0.000	1.000 **(Paradoxical)**	0.000
Murphy Method [cite: 14]	0.499	0.499	0.002	0.000
Sun Method [cite: 14]	0.182	0.182	0.003	0.633 **(Ignorance)**
PCR5 / PCR6 Rule [cite: 5]	0.494	0.494	0.012	0.000

The PCR5/6 rules isolate the conflict product ($m_1(R)m_2(W) = 0.9801$) and redistribute it back to R and W proportionally to their original masses[4]. This maintains the true diagnostic uncertainty, signaling to the downstream controller that further evidence is required[4].

In machine learning and domain adaptation, this conflict can be extracted explicitly. By constructing a cross-domain transition matrix M, where the trace of the matrix represents prediction correlation, researchers have successfully quantified "environmental disagreement"[16]. When non-causal environment features disagree, this trace acts as an active discriminator feature to guide the model to ignore misleading variables and avoid negative transfer[16].

Confidence

High. The mathematical limitations of classical DST and the robustness of PCR5/6 rules under high contradiction are well-established in multi-sensor fusion theory[5].

Data gaps and domain-transfer caveats

While the mathematics of PCR5/PCR6 is highly mature, empirical implementation within automotive engine control units (ECUs) is extremely rare. Legacy vehicle diagnostics (1990–2010) are severely limited by computational constraints (8-bit to 32-bit microcontrollers with low memory limits). The localized nature of PCR calculations (which require storing cross-products of beliefs for combinations of hypotheses) represents a major structural barrier for on-vehicle embedded systems[5]. However, this is not a constraint for off-board PC-based garage diagnostic scan tools.

Calibrated-likelihood / Bayesian fusion vs additive weights

The third research question evaluates how calibrated-likelihood and Bayesian fusion approaches compare with simple additive weighting when evidence channels are conditionally dependent, and how to prevent weak channels from being swamped.

Finding

The naive Bayes conditional independence assumption—which assumes that the joint probability of the evidence is the product of individual channel probabilities—is highly destructive when channels are correlated[19]. In the engine under study, volumetric efficiency cross-checks, calculated engine load, and short/long-term fuel trims are highly correlated physical responses to a single root cause (e.g., an intake manifold vacuum leak).

Under naive independence assumptions, these correlated features cause "swamping," where the overlapping signals inflate the joint probability of a specific family, completely drowning out independent diagnostic channels (e.g., raw gas chemistry)[19].

To prevent swamping while preserving diagnostic performance under missing-evidence states (such as the absence of a DTC), three architectural corrections are validated[6]:

1. **Bayesian Belief Networks (BBNs)**: Directed Acyclic Graphs (DAGs) that explicitly model the causal dependencies between sensors[6].
2. **Stacked Generalization (Stacking over channels)**: Using logistic regression or a Log-Odds Precedent Aggregator (LOPA) to aggregate channel-level predictions, which naturally prunes redundant dependencies[7].
3. **Leak Probabilities in Noisy-OR structures**: Allocating a non-zero probability to unmodeled causes to prevent the network from collapsing to zero when evidence channels are starved[20].

Evidence and sources

In complex engine systems like diesel and petrol powertrains, research shows that a pure model-based or data-driven diagnostic network often fails under variable loads or when sensors degrade[21].

To address this, a "Belief Rule-Based" (BRB) expert system was developed for marine diesel engines, which integrates expert-elicited qualitative rules with numerical sensor data using Dempster-Shafer evidential reasoning[22]. This hybrid model was benchmarked against Artificial

Neural Networks (ANN), Support Vector Machines (SVM), and binary logistic regression under five-fold cross-validation[22]:

- **Diagnostic Performance on Engine Wear**: The optimized BRB model achieved a top-1 diagnostic accuracy of 95.56%[24]. It consistently outperformed the standalone ANN, SVM, and binary logistic regression models in terms of diagnostic stability, computational footprint, and multi-fault identification[23].
- **Handling Starved Inputs**: By incorporating a structural **"leak probability"** parameter—representing the background probability of unmodeled causes—the model maintained stable classifications even when input indicators were missing[20].

Furthermore, in multi-DTC vehicle electronic systems, incorporating a causal BBN rather than standard decision trees dramatically reduced misdiagnoses[6]. Under standard rule-based structures, if any intermediate sensor in the logical chain fails or is absent, the system misdiagnoses or stalls[6].

The BBN models the joint probability $P(\text{Fault} \mid \text{Symptom}_1, \ldots, \text{Symptom}_n)$ and can reversely infer the most likely failure cause even when multiple DTCs are missing or logged simultaneously[6].

To mitigate feature correlation and swamping, the Log-Odds Precedent Aggregator (LOPA) replaces standard naive Bayes product fusion with a log-odds aggregation of per-source reliability estimates[7]. LOPA mathematically prunes redundant chains and prevents highly correlated symptoms from compounding and overwhelming the independent diagnostic channels[7].

Confidence

High. The mathematical consequences of conditional independence violations (such as the XOR limitation and joint probability inflation) are proven statistical realities[19]. BBN implementations in complex engine platforms show consistent performance gains[6].

Data gaps and domain-transfer caveats

The high-performing BRB and BBN engine benchmarks cited are predominantly conducted on marine diesel powertrains[22] or simulation models (e.g., CWRU rolling bearing databases)[26]. These systems exhibit slower dynamic transients than passenger car petrol engines, which must continuously adapt to fast-changing driver demands and closed-loop three-way catalyst behaviors.

Furthermore, parameterizing the CPTs of a large Bayesian Network requires highly specialized software tools (such as GeNIe or SMILE) and significant historical datasets[27], which are rarely standardized across legacy automotive service networks.

Method comparison table

The following table compares the diagnostic fusion paradigms evaluated in this report, detailing their advantages, structural vulnerabilities, comparable performance, and overall

architectural maturity:

Approach	What it Buys	What it Costs / When it Fails	Evidence It Beats Additive Weighting on Comparable Tasks	Maturity
Additive Weights *(Current Paradigm)*	Extreme simplicity; easily written in basic logic; requires no complex matrix calculations; highly interpretable at a single-rule level.	Linear constraints; cannot model feature dependencies; prone to "outranking" errors; completely fails under input starvation[3].	Evaluated as the baseline. Plateaus at $50–52\%$ top-1 accuracy in the legacy petrol passenger car system under study.	Legacy (1980s–1990s standard).
Dempster-Shafer Conflict-Aware (PCR5/6)	Explicitly quantifies and isolates channel contradictions; stops single-sensor overrides of dominant signals[4].	Highly complex combinatorial calculations; requires defined basic belief assignments for all subsets[4].	Replaces Dempster's rule under high conflict ($K =$). Corrects the counter-intuitive BBA distribution from 100% error to a balanced distribution[14].	Research grade; high usage in military and advanced multi-sensor tracking.
Bayesian Network (BBN / TAN)	Models true causal dependencies (DAGs)[6]; provides	Constructing and maintaining the Directed Acyclic Graph	On the HEPAR medical dataset, the BN (Hepar-BN) achieved	Mature in academic and aerospace health management;

	calibrated probabilities; naturally handles missing DTCs or sensors[28].	structure is highly labor-intensive ; CPTs are difficult to populate[30].	94.0% sensitivity compared to only 62.0% for the rule-based model[3].	emerging in advanced automotive workshop tools[6].
Stacked Logistic Regression / LOPA	Merges channels using learned, robust weights; prevents redundant, highly correlated sensors from swamping[7].	Requires a centralized, labeled training dataset of unified sensor/gas cases to fit stacking coefficients.	Outperforms simple majority voting and naive Bayes by pruning redundant diagnostic chains in legal and ML benchmark domains[7].	High industrial maturity in general machine learning; low integration in embedded controllers.
Learned Rankers (Random Forest / XGBoost)	Exceptional multi-class separation; handles hundreds of raw variables; completely automates feature engineering[8].	Acts as a complete "black box"; cannot provide physical or logical explanations; highly sensitive to training data shift[33].	On simulated electrical grid and mechanical fault datasets, Random Forests achieved 88.17% accuracy compared to 79.01% for standard Naive Bayes[8].	High industrial maturity; standard for cloud-based predictive maintenance.

Consolidated source table

This table aggregates the peer-reviewed literature, dataset evaluations, and technical standards supporting the assertions and findings in this report:

Source	Type	Year	Domain	Claim it Supports	Access	Link
Onisko et al.	Peer-Reviewed Journal	2001	Medical / Diagnostics	HEPAR study results; rules plateau at 62% sensitivity whereas Bayesian networks reach 94%[3].	Free	AIME 2001 Paper
Aragao & Alcantara	Technical Preprint	2024	Information Fusion	DS combination rule anomalies; Murphy and PCR5/6 conflict redistribution evaluation[14].	Free	arXiv:2408.08928
Amin & Karmilasari	Peer-Reviewed Journal	2021	Automotive Diagnostics	Standard DSmT vs. DST comparison; managing high conflict scenarios[4].	Free	IRJAES Journal PDF

He et al.	Peer-Reviewed Journal	2021	Power Systems / ML	Random Forest (88.17%) outperforming Naive Bayes (79.01%) on multi-class fault diagnostics[8].	Free	Frontiers in Energy Research
Xu et al.	Peer-Reviewed Journal	2017	Powertrain / Marine	Belief Rule-Based expert systems reaching 95.56% accuracy and outperforming standard ANNs and SVMs[24].	Paid	IEEE SMC Paper
Dezert & Smarandache	Monograph	2006	Sensor Fusion	PCR5/6 mathematical formulas; proportional conflict redistribution rules to resolve Zadeh's	Free	UNM Math Monograph

				paradox[5].		
Bings et al.	Conference Proceeding	2025	Legal Reasoning / AI	LOPA framework; aggregating precedent opinions in posterior log-odds to prune redundant chains[7].	Free	JURIX 2025 LOPA
Hu et al.	Review Paper	2018	Chemistry / Machine Learning	High-dimensional swamping of Naive Bayes signals by correlated, non-relevant features[19].	Free	ACS Chemical Reviews
Shao et al.	Peer-Reviewed Journal	2022	Industrial Process Control	Certainty factors vs. LSTM networks on aluminum cell diagnostics (94.2% accuracy)	Free	MDPI Processes Journal

				[2].		
Xu & Yang	Peer-Reviewed Monograph	2021	Automotive / Electronics	Bayesian Belief Network (BBN) framework for multi-DTC vehicle electronic system troubleshooting[6].	Paid	ResearchGate Publication
Li et al.	Peer-Reviewed Journal	2025	Domain Adaptation / ML	Disagreement and transition matrices M to model non-causal environmental conflict[17].	Free	arXiv:2510.24044v1

Could not be found list

The following gaps in empirical evidence have been identified in the literature search:

1. **Direct Head-to-Head Comparative Benchmarks for Legacy Petrol Fleets**: No publicly documented, peer-reviewed datasets exist that directly compare a hand-tuned additive rule-based engine against a Bayesian Network or Random Forest using the four exact diagnostic channels under study (5-gas exhaust chemistry, live OBD-II sensor streams, generic P-codes, and operator symptoms) for model years 1990–2010.
2. **Computational Benchmarks for PCR5/6 on Embedded Legacy ECUs**: Quantitative data documenting the execution time and memory footprint of Proportional Conflict Redistribution rules (PCR5 or PCR6) compiled on 8-bit, 16-bit, or early 32-bit automotive microcontrollers (e.g., Motorola 68HC11 or Infineon TriCore) is absent.
3. **Statistical Characterization of Operator-Reported Symptom Noise**: No peer-reviewed studies quantify the exact prior probability distributions, error rates, or likelihood noise for qualitative operator complaints (e.g., "runs rough") when fused with quantitative combustion chemistry datasets in real-time diagnostics.

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